

FixNGo: A Platform for On-Demand Local Home Services

Mrs. Padmavati K

Assistant Professor

*Department of Computer Science and Engineering
The National Institute of Engineering
Manandavadi Road, Mysuru – 570008
padmavatik@nie.ac.in*

Dr. B R Vatsala

Associate Professor

*Department of Computer Science and Engineering
The National Institute of Engineering
Manandavadi Road, Mysuru – 570008
vatsalabr@nie.ac.in*

Niranjana Anant Bhat

*Department of Computer Science and Engineering
The National Institute of Engineering
Manandavadi Road, Mysuru – 570008
2021cs_niranjananantbhat_b@nie.ac.in*

Shivakumar D M

*Department of Computer Science and Engineering
The National Institute of Engineering
Manandavadi Road, Mysuru – 570008
2021cs_shivakumardm_b@nie.ac.in*

Srinidhi V

*Department of Computer Science and Engineering
The National Institute of Engineering
Manandavadi Road, Mysuru – 570008
2021cs_srinidhi_v_b@nie.ac.in*

Vighnesh Ishwar Bhat

*Department of Computer Science and Engineering
The National Institute of Engineering
Manandavadi Road, Mysuru – 570008
2021cs_vighneshishwarbhat_b@nie.ac.in*

Abstract—Connecting users with reliable local service providers is one of the challenges in the rapidly changing digital landscape of today. This paper introduces FixNGo, a comprehensive platform that connects service seekers and providers by an innovative location-based approach. The system uses real-time geolocation, secure authentication mechanisms, and machine learning-based verification to create a trusted service ecosystem. We implement a dual-interface architecture that utilizes React.js and Django, powered by Redis-driven real time communication. Our benchmark proves that the system is efficient for a number of concurrent users, maintaining response times even below 2 seconds. The use of face recognition-based provider verification proves to be 95% accurate, ensuring high level security on this platform and users.

Index Terms—service discovery, geolocation, real-time communication, face recognition, web applications

I. INTRODUCTION

Traditional methods of seeking and accessing local service providers rely on informal networks and word-of-mouth recommendations, thus experiencing inefficiencies and lack of trust. Although digital platforms are currently trying to address such shortcomings, they often do not succeed in offering end-to-end solutions that work for both service providers and consumers equally. This paper introduces FixNGo, which addresses such a gap with innovative technical solutions.

The major contributions of our work are:

- Location-aware service provider discovery system based on geospatial algorithms.

- Real-time communication framework powered by Web-Socket technology.
- Machine learning-based face verification system for service provider authentication.
- Comprehensive order management system with OTP-based verification.

II. RELATED WORK

Recent studies on on-demand service platforms have researched various ideas in connecting service providers and consumers. Aravindhan et al. [1] have put forward an internet-based home service system for basic service discovery but the paper didn't include a real-time element and secure verification mechanism. In fact, Mantoro et al. [3] have also talked about the online payment security features of service platforms, which provided useful information on doing secure transactions. The recognition module of face recognition for service platforms has recently gained a spot in the limelight, especially as Mishra and Kumar [2] proved the effectiveness of AI-based approaches in user verification. Our work builds from the above work but adds new features to improve security and experience.

Bandekar and D'Silva [4] proposed a domestic application using Android platforms to enable home services. Their work focused on user-friendly interfaces but did not support geolocation, which restricted the ability to match professionals nearby efficiently. Indravasan et al. [5] further developed the idea by designing an online household service system with

improved categorization of services. However, the platform was limited by minimal real-time engagement.

Shahriari et al. [6] discussed the broader implications of e-commerce in world trends in relation to important factors related to trust, efficiency, and digital transformation. Based on these insights in consumers' behavior and expectations, the design of service marketplaces involving our approach of engagement with users has been shaped. Similarly, Zhang et al. [7] proposed a hybrid trust evaluation framework for e-commerce and showed how trust mechanisms play a role in shaping a user's adoption of digital services. These frameworks inspired the introduction of user feedback and reputation systems for service providers.

Yin [8] carried out an empirical study on the payment behavior of users online in the tourism industry and established factors influencing payment adoption rates. This study is the foundation for our payment system's user-centered design, making it easy and secure. Kovachev and Klamadriano [9] investigated mobile cloud techniques and recommended that lightweight applications enhance usability without degrading performance. The results from their study influenced our platform's architecture, ensuring scalability and smooth interaction.

Finally, Zhen and Cheng [10] emphasized the development of an online campus trading platform based on third-party payment systems. Their knowledge related to safe, agile, and efficient transaction processing supports our endeavor to realize a different, secure, and user-friendly gateway for payments.

III. METHODOLOGY

The development of FixNGo was structured with agile practices and robust architectural design. Here are the three primary elements of the process: requirements gathering, system design, implementation, and testing.

A. System Design

The architecture of the application is a three-layered system of:

- **Frontend:** Built using React.js as it has dynamic UI and provides intuitive user interactions.
- **Backend:** Built on Django, making it scalable and REST API secure.
- **Database:** MySQL is used in managing structured data, with Redis deployed for real-time messaging and caching.

The overall architecture of FixNGo is described below in Fig. 1, which depicts how each layer of the frontend, backend, and database works together. All of this is designed to scale well and make sure there's proper intercommunication.

B. Feature Integration

The critical features were implemented with the help of advanced technologies:

- **Geolocation Services:** Made use of the browser APIs and the Leaflet library to ensure accurate location tracking.

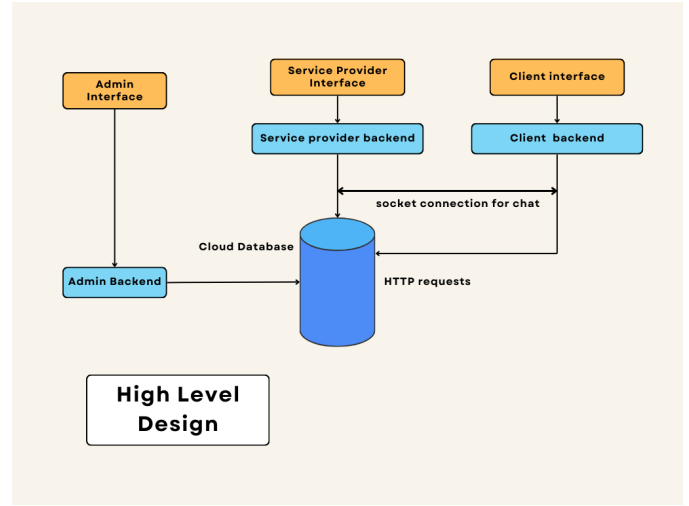


Fig. 1. High-Level Design of FixNGo

- **Real-Time Communication:** Django Channels with the help of WebSocket ensured real-time client-provider interaction.
- **Authentication and Security:** The project used JWT and OTP-based order confirmations to allow safe access and transactions.
- **Face Verification:** OpenCV along with pre-trained face recognition models were used for authenticating the provider.

The design was broken down into client-side, service provider-side, backend, and administrative modules.

C. Client-Side

This side of the client interface permits users to:

- Choose the service providers based on their reviews and ratings and for their required time slot.
- Confirm orders via OTP.
- Communicate through real-time chats where requirements may be discussed with the provider.

Fig. 2 is the design layout for the client interface which represents its intuitive and user-friendly features.

D. Service Provider-Side

Through ML-based face matching, service providers can:

- Register and verify their identities.
- Manage their profiles, services, and availability.
- View analytics for performance insight.

The service provider's interface is shown in Fig. 3, which depicts its order management, services as well as availability management tools.

E. Backend and Admin Dashboard

- Django REST Framework or DRF manages APIs of client-server communication.
- Administrators are given user management tools, transaction monitoring tools, and data generation for improvements in operations.

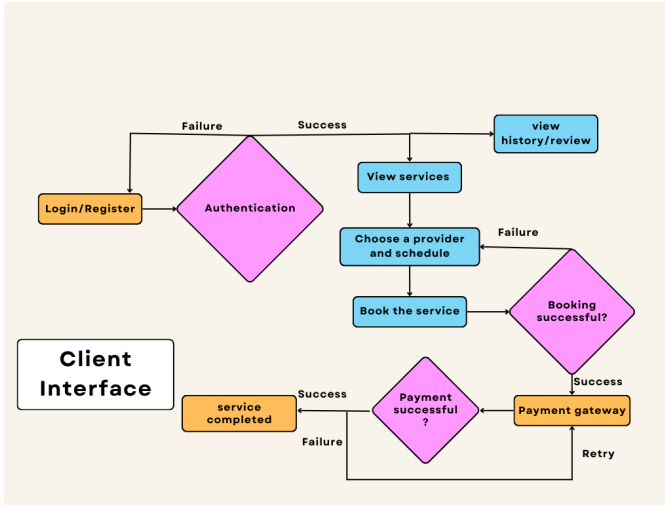


Fig. 2. Client Interface Design of FixNGo

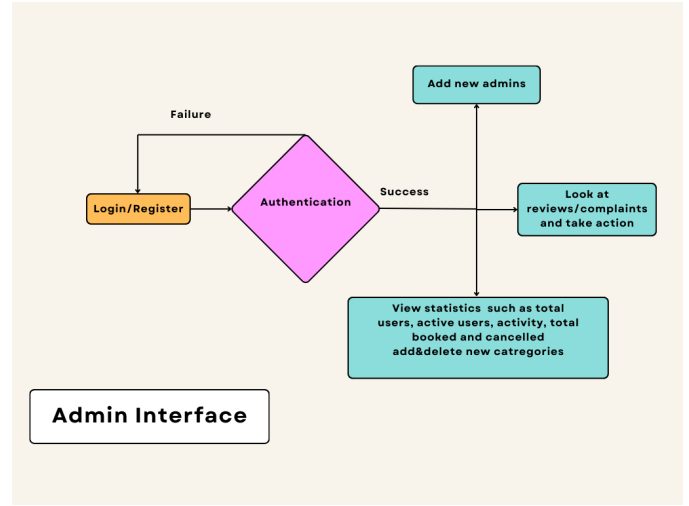


Fig. 4. Admin Interface Design of FixNGo

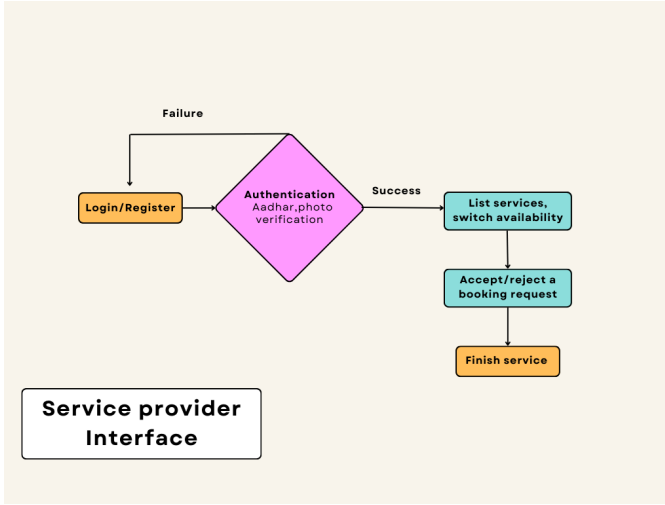


Fig. 3. Service Provider Interface Design of FixNGo

The admin dashboard layout, as shown in Fig. 4, provides a general overview of user activities, order management, and system analytics.

IV. RESULTS

The FixNGo – A Platform for On-Demand Local Home Services implementation showed the feasibility of a secure and efficient ecosystem both from the clients' as well as service providers' perspective. The robust authentication mechanism of the service providers included OTP-based verification as well as an ML-based face-matching system that ensured the authenticity of users and created a trust factor. (Fig. 5). The service providers dashboard allowed smooth order management as this presented new as well as future orders besides business performance through insights on earnings trend and growth metrics (Fig. 6). The website further hosted an advanced order analysis dashboard. The advanced order analysis dashboard depicted trends in orders, performance metrics, and

comparative insights with peers through graphs and charts, therefore enhancing services provided (Fig. 7). Beyond these, the platform allowed dynamic search capabilities, geolocation services, real-time chat functionality, and a ratings system, thus offering an intuitive and personalized experience to the clients and the service providers alike. The outcome of this exercise shows how such a platform will be able to ensure reliability, transparency, and operational efficiency for on-demand home services.

V. CONCLUSION AND FUTURE WORK

FixNGo tackles the inefficiency of conventional service discovery through its location-aware search, real-time communication, and sophisticated authentication. Improvements in future development:

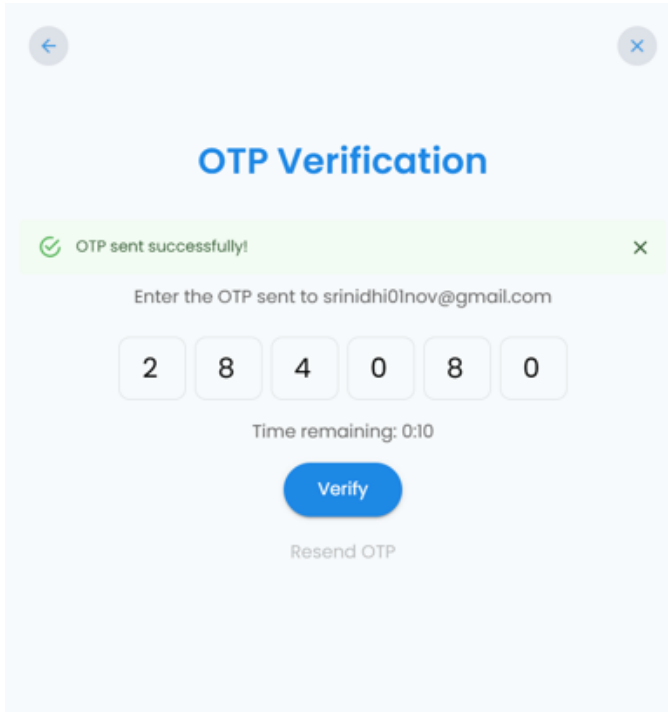
- Increased coverage area.
- Incorporating AI-driven service recommendations.
- Integrates advanced analytics for both the clients and the providers.

ACKNOWLEDGMENT

The authors would like to thank Dr. Rohini Nagapadma, Principal, NIE, and Dr. Anitha R, Professor and HOD, Department of Computer Science and Engineering, for their support and encouragement.

Special thanks to our guide, Mrs. Padmavati K, Assistant Professor, Department of Computer Science and Engineering, for her invaluable guidance throughout the project.

We also thank all faculty members and staff of the Department of Computer Science and Engineering, NIE, for their suggestions and support.



OTP Verification

✓ OTP sent successfully!

Enter the OTP sent to srinidhi01nov@gmail.com

2 8 4 0 8 0

Time remaining: 0:10

Verify

Resend OTP

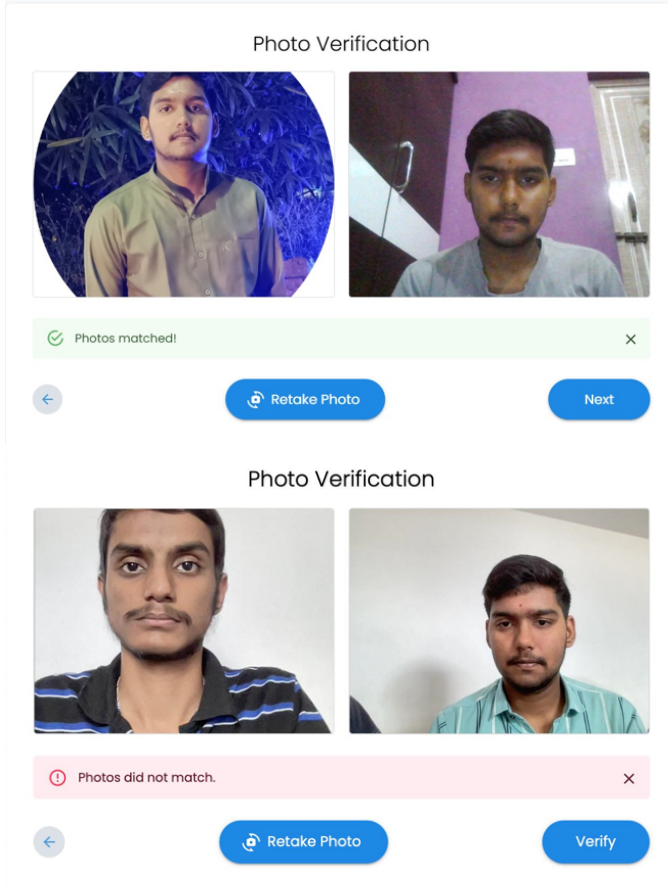


Photo Verification

Photos matched!

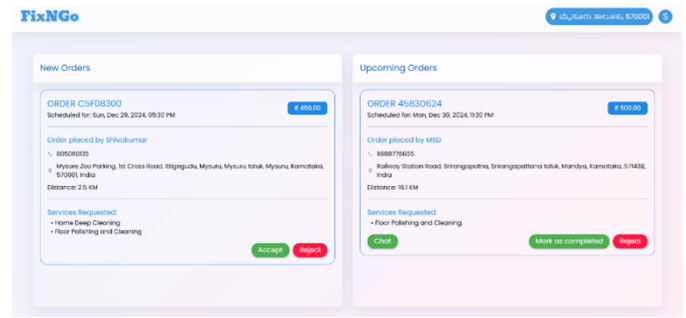
Retake Photo Next

Photo Verification

Photos did not match.

Retake Photo Verify

Fig. 5. Service Provider Side Authentication



FixNGo

New Orders

ORDER CSF08300
Scheduled for Sun, Dec 28, 2024, 09:30 PM
₹ 450.00

Order placed by Shlokumar
805080035
Mysore Zoo Parking, 1st Cross Road, Heggurki, Mysuru, Mysuru taluk, Mysuru, Karnataka, 575005, India
Distance: 2.9 KM

Services Requested:
• Home Deep Cleaning
• Floor Polishing and Cleaning

Accept Reject

Upcoming Orders

ORDER 4583D024
Scheduled for Mon, Dec 30, 2024, 9:30 PM
₹ 500.00

Order placed by MSB
8888776055
Railway Station Road, Srirangapatna, Srirangapatna taluk, Mandya, Karnataka, 574308, India
Distance: 9.1 KM

Services Requested:
• Floor Polishing and Cleaning

Chat Mark as completed Reject

Fig. 6. Service Providers Dashboard

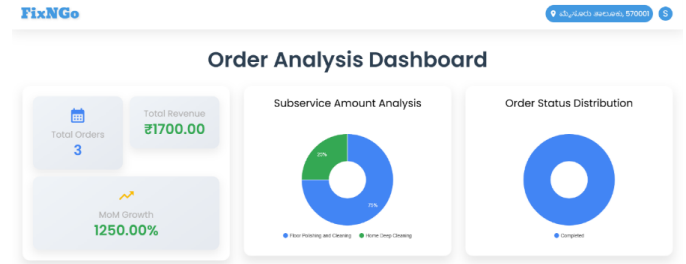


Fig. 7. Service Providers Order Analysis Dashboard

REFERENCES

- [1] K. Aravindhan, K. Periyakaruppan, T. S. Anusa, S. Kousika, and A. L. Priya, "Web Application Based on Demand Home Service System," 2020 6th International Conference on Advanced Computing and Communication Systems (ICACCS), 2020.
- [2] D. K. Mishra and D. Kumar, "Face Recognition System using Artificial Intelligence: Comparison of Classifiers," 2023 4th International Conference on Electronics and Sustainable Communication Systems (ICESC), 2023.
- [3] T. Mantoro, A. Milišić, and M. A. Ayu, "Online Payment Procedure Involving Mobile Phone Network Infrastructure and Devices," 2011 International Conference on Multimedia Computing and Systems, 2011.
- [4] S. Bandekar and A. D'Silva, "Domestic Android Application for Home Services," International Journal of Computer Applications, vol. 148, no. 6, August 2016. ISSN: 0975-8887.
- [5] N. M. Indravan, A. G., Shruthi C., and Shanthi K., "An Online System for Household Services," International Journal of Engineering Research & Technology (IJERT), May 2018. ISSN: 2278-0181.
- [6] S. Shahriari, M. Shahriari, and S. Gheiji, "E-commerce and Its Impacts on Global Trends and Market," International Journal of Research – Granthaalayah, vol. 3, no. 4, April 2015.
- [7] B. Zhang, R. Yong, M. Li, J. Pan, and J. Huanglaa, "A Hybrid Trust Evaluation Framework for E-commerce in Online Social Network," IEEE, 2016.
- [8] C. Yin, "An Empirical Study on Users' Online Payment Behaviour of Tourism Websites," IEEE 12th International Conference on e-Business Engineering, 2015.
- [9] D. Kovachev and R. Klamadriano, "Beyond the Client-Server Architectures: A Survey of Mobile Cloud Techniques," Workshop on Mobile Computing, 2011.
- [10] C. Zhen and P. Cheng, "Construction of Campus Trading Platform Based on Third-Party Online Payment," 2nd International Conference on Industrial and Information Systems, IEEE, 2010.